AMD DEVELOPER HACKATHON: ACT II · VIDEO CAPTIONING TRACK

StyleCap

One video in. **Four voices out.**

Formal

Sarcastic

Humorous · tech

Humorous · everyday

github.com/Ov3rkill15/amd-video-captioning

01 The task

Track 2 is harness-judged: the container receives video URLs and must return one caption per requested style, faithful to the footage and true to the tone. An LLM judge scores every caption on **accuracy** and **style match**.

`/input/tasks.json`

```
{ "task_id": "1",
  "video_url": "https://...clip.mp4",
  "styles": ["formal", "sarcastic",
    "humorous_tech", "humorous_non_tech"] }
```

`/output/results.json`

```
{ "task_id": "1",
  "captions": {
    "formal": "...", "sarcastic": "...",
    "humorous_tech": "...", ... } }
```

Hidden evaluation set: ~12 unseen clips, 30s to 2min, spanning nature, urban, animals, people, sports, food, weather, and technology.

02 How it works

STEP 1

Download

Fetch each clip; 4 tasks run in parallel workers.

STEP 2

Sample frames

8 evenly spaced frames via OpenCV, in memory, downscaled to 768px.

STEP 3

One VLM call

A single multimodal request to Fireworks AI (kimi-k2p6) returns all four styles at once.

STEP 4

Validate & write

Robust JSON parsing, then exact harness-format output.

Tone quality is engineered, not hoped for

Few-shot tone exemplars are baked into the prompt. They were selected with our own LLM-judge evaluation harness (`scripts/eval_captions.py`), which scores candidate prompts on the same two dimensions the judges use.

03 Built for the harness

~20s

3 official UHD clips end to end (10 min budget)

1 call

per video, all four styles at once

1.21 GB

linux/amd64 image (10 GB limit)

→ Exit code 0, valid JSON, every requested style always present

→ Public image: `ghcr.io/ov3rkill15/amd-video-captioning:latest`, anonymously pullable

→ Container starts instantly (no model weights inside; inference is serverless via API)

04 A broken clip never sinks the run

Three fallback layers guarantee a complete, valid answer for every task:

LAYER 1

API retries

Transient errors and rate limits are retried with backoff.

LAYER 2

Per-style recovery

If the combined call fails to parse, each style is requested individually.

LAYER 3

Guaranteed captions

As a last resort, safe generic captions are emitted, so no style is ever missing.

Missing styles score zero for a clip. With this design, that failure mode is structurally impossible: the container always exits 0 with a complete `results.json`.

05 Measured, not guessed

Our eval harness replays the judges' rubric: an LLM judge scores **accuracy** and **style** over 6 diverse clips × 4 styles, for every pipeline variant we considered.

0.81

shipped: kimi-k2p6 + tone exemplars

0.80

self-critique pass · 0.79 best-of-3 · 0.78 two-stage

0.77

Gemma 4 26B A/B via OpenRouter · 0.70 with 16 frames

- → The shipped config beat every challenger, including test-time compute (best-of-3) and a self-critique pass
- → Counter-intuitive finding: doubling frames 8 → 16 **hurt** accuracy (0.81 → 0.70)
- → Raw scored outputs are committed under `demo/samples/eval_*.json`

06 Sample output · official kitten clip

FORMAL

A small orange kitten walks slowly toward the camera through a shaded path bordered by dense green foliage and tree trunks.

SARCASTIC

Breaking news: a kitten walks forward. Nature documentaries will never recover from this level of drama.

HUMOROUS · TECH

Kernel panic averted. This orange fluff successfully compiled a walk.exe with zero segmentation faults.

HUMOROUS · EVERYDAY

This kitten is walking toward you like it already knows you have no treats and does not care.

07 Generalisation, not memorisation

→ Prompts are purely style-driven; nothing is tuned to the example clips

→ Same pipeline handles nature, urban, animals, people, sports, food, weather, tech footage

→ Provider-agnostic client (OpenAI-compatible), model swappable via env config

→ Modular repo: `src/pipeline` · `src/models` · `scripts/eval_captions.py` · local web demo UI

Image

`ghcr.io/ov3rkill15/amd-video-captioning:latest`

Source

`github.com/Ov3rkill15/amd-video-captioning`